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KINEMATICS

Kine ⇒ Motion , Matics ⇒ Maths.

⇒ MOTION IN ONE DIMENSION.
(linear or Rectilinear).

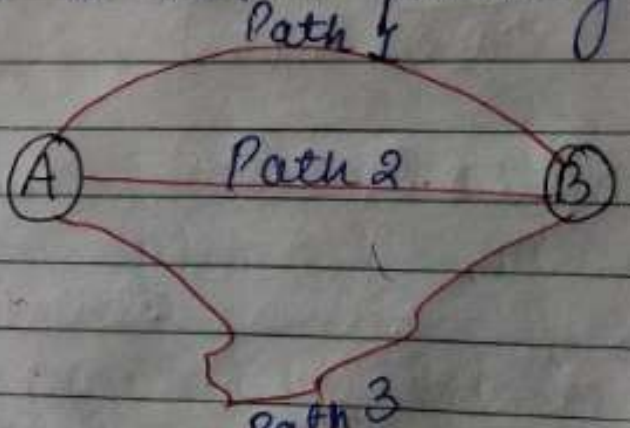
VARIABLE

* Distance
* Displacement
* Velocity
* Acceleration
* Speed
* Jerk

DISTANCE AND DISPLACEMENT

* Distance

⇒ It is a scaler quantity



Path 1, Path 2, Path 3 are the distance between A and B.

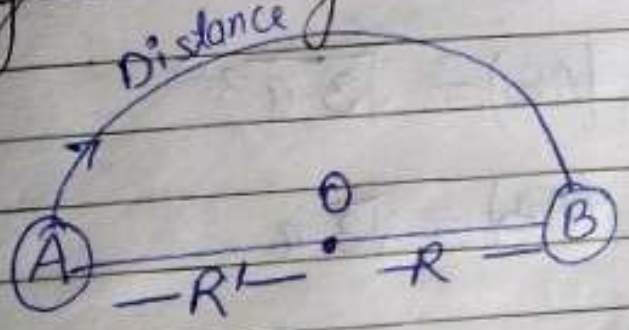
* Displacement.
⇒ It is a vector quantity

Path 2 is the displacement. (It is the shortest distance from initial position to final position.

Relation between Distance and displacement.

(1) Distance > displacement
(2) Distance = displacement
(3) Distance ≥ displacement
(4) Distance < displacement

Ques- Find the ratio of distance to displacement in the following diagram. (The given diagram is semi circle)



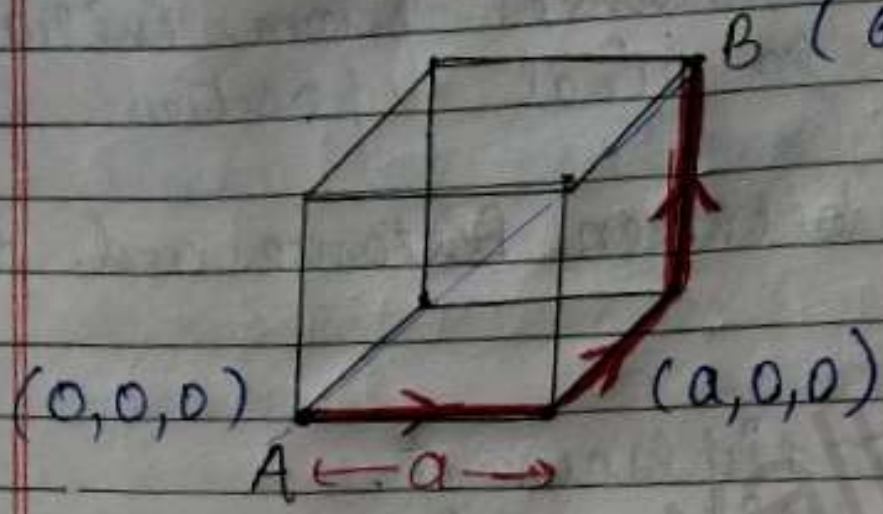
Ratio of $\frac{\text{distance}}{\text{displacement}} = \frac{\pi R}{2R} = \frac{\pi}{2}$

"No religion has mandated killing others as a requirement for its sustenance or promotion." —Dr.A.P.J.Abdul Kalam

Date / /

cube.

Ques6- Consider a square, where distance of a bee is shown in figure. Find the shortest distance between A and B.



Shortest distance of a cube is AB (body diagonal)

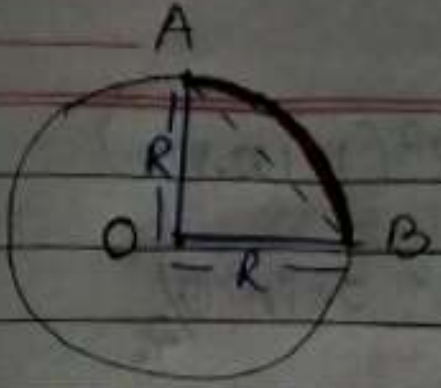
Shortest distance $AB = a\hat{i} + a\hat{j} - a\hat{k}$

$$|AB| = \sqrt{a^2 + a^2 + a^2}$$

$$|AB| = \sqrt{3a^2}$$

$$|AB| = \sqrt{3}\,a.$$

Ques6- A particle covers $\frac{1}{4}$th of circle, find the displacement of particle



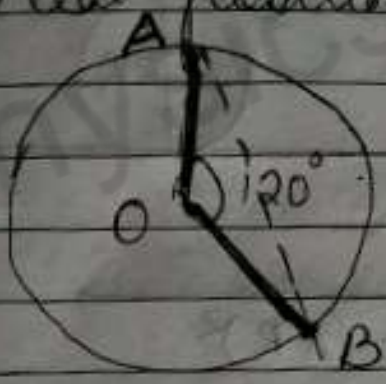
$(AB)^2 = (OA)^2 + (OB)^2$

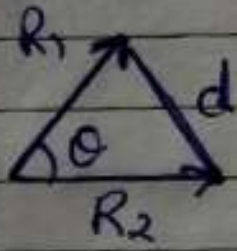
$(AB)^2 = R^2 + R^2$

$(AB)^2 = 2R^2$

$AB = \sqrt{2}R$.

Ques6 - Find the displacement of particle, if particle makes 120° angle with its initial position.





We know that

$\vec{R_1} + \vec{d} = \vec{R_2}$

$\vec{d} = \vec{R_2} - \vec{R_1}$

$|\vec{d}| = \sqrt{R_1^2 + R_2^2 - 2R_1R_2\cos\theta}$

$|d| = \sqrt{R^2 + R^2 - 2R^2\cos\theta}$

"We authors are not against any policies of government for its sustenance or promotion." —Dr. A.P.J. Abdul Kalam

Date ___/___/_____

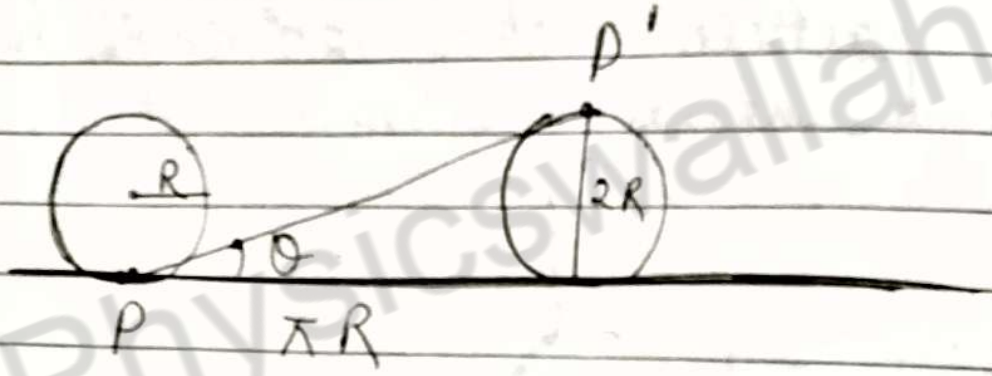
$$|d| = \sqrt{2R^2(1-\cos\theta)}$$

$1-\cos\theta = 2\sin^2\theta/2$

$$|d| = \sqrt{2R^2 \, 2\sin^2\theta/2}$$

$$|d| = 2R\sin\theta/2$$

Ques 6- A wheel of radius R is rolling on a road, what will be the displacement of a point P in half revolving of wheel on road.



$$PP' = \sqrt{(2R)^2 + (\pi R)^2}$$

$$PP' = \sqrt{4R^2 + \pi^2 R^2}$$

$$PP' = \sqrt{R^2(4+\pi^2)}$$

$$PP' = R\sqrt{4+\pi^2}$$

Direction:

$$\tan\theta = \frac{P}{B} = \frac{2R}{\pi R} = \frac{2}{\pi}$$

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SPEED AND VELOCITY

* SPEED

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Speed is a scalar quantity.

⇒ Average speed
⇒ Instantaneous speed (At particular instant (dt))

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

Average speed is for large time (बड़े समय के लिए) (Δt)

* VELOCITY

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

Velocity is a vector quantity

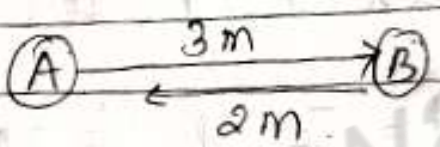
⇒ Average velocity (Δt)
⇒ Instantaneous velocity (At particular instant (dt))

Date __/__/____

Relation between average speed and average velocity.

Average speed ≥ average velocity

Ques- A boy walks 3m east in 2 sec. then 2m west in 3 sec. find his
⇒ Average speed.
⇒ Average velocity.



Average speed = $\frac{\text{Total Distance}}{\text{Total time}}$

$= \frac{3+2}{5} = 1 \text{ m/sec}.$

Average velocity = $\frac{\text{Total Displacement}}{\text{Total time}}$

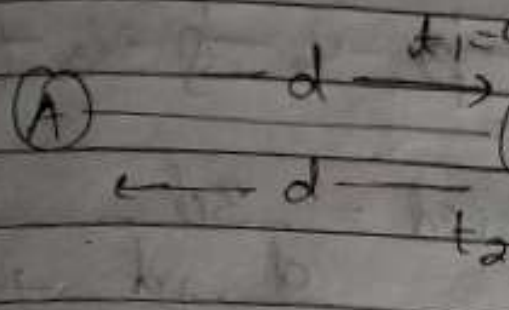
$= \frac{1}{5} = 0.2 \text{ m/sec}$

Ques- A man goes from A to B at v_1 speed and returns from B to A at v_2 speed find
(1) Average velocity.
(2) average speed

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Ans (i) Average velocity = $\frac{0}{t}$ = 0 m/sec.

(ii) average speed = $\frac{\text{total distance}}{\text{total time}}$



$$\text{av. speed} = \frac{2d}{\frac{d}{v_1} + \frac{d}{v_2}}$$

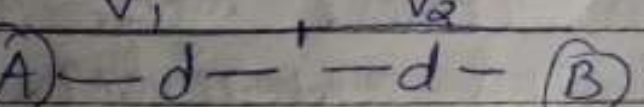
$$\text{av. speed} = \frac{2d}{\frac{dv_2 + dv_1}{v_1 v_2}}$$

$$\text{av. speed} = \frac{2v_1v_2}{v_1 + v_2}$$

Ques- A man covers half distance at v_1 speed and half at v_2 speed. find average speed.

$t_1 = d/v_1$ $\quad t_2 = d/v_2$

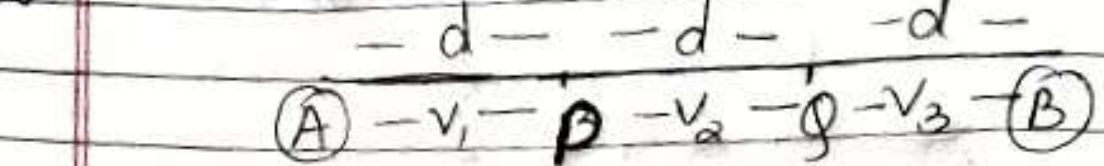
Ans-



$$\text{Av speed} = \frac{2d}{t_1 + t_2}$$

Ques - A man covers 3 equal distance with speed v_1, v_2, v_3. Find his average speed.

Ans -



$$\text{Average speed} = \frac{3d}{\frac{d}{v_1} + \frac{d}{v_2} + \frac{d}{v_3}}$$

$$\text{av. speed} = \frac{3}{\frac{1}{v_1} + \frac{1}{v_2} + \frac{1}{v_3}}$$

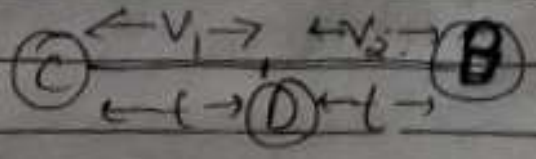
$$\text{av. speed} = \frac{3}{\frac{v_2v_3 + v_1v_3 + v_1v_2}{v_1v_2v_3}}$$

$$\text{average speed} = \frac{3v_1v_2v_3}{v_1v_2 + v_2v_3 + v_3v_1}$$

Ques - A man covers half distance with speed v and remaining half with speed v_1 and v_2 for equal time interval. Find his average speed.

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$$2t \Rightarrow 2\left(\frac{d}{v_1+v_2}\right)$$

$$t = \frac{d}{v_1+v_2}$$

$$\text{Av. speed} = \frac{2d}{\frac{d}{v} + \frac{2d}{v_1+v_2}}$$

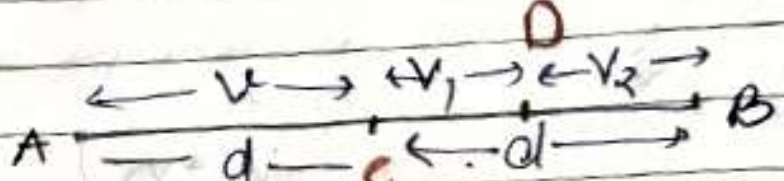
$$= \frac{2}{\frac{1}{v} + \frac{2}{v_1+v_2}}$$

$$= \frac{2}{\frac{v_1+v_2+2v}{v(v_1+v_2)}}$$

$$\text{Av. speed} = \frac{2v(v_1+v_2)}{v_1+v_2+2v}$$

Ques:- A bus covers half distance at speed v and remaining half at speed v_1 for $\frac{1}{3}$ time and v_2 for $\frac{2}{3}$ time. find average speed.

Ans:-



For AC, $t_1 = \frac{d}{v}$

for CD, $d = s \times t$

$d = v_1 \times \frac{t}{3}$

for DB, $d = s \times t$

$d = v_2 \times \frac{2t}{3}$

$d = v_2 \times \frac{2t}{3}$

Total from C to B

$$d = \frac{v_1 t}{3} + \frac{3}{3} v_2 t$$

$$t = \frac{3d}{v_1 + 2v_2}$$

Av. velocity $= \frac{2d}{\frac{d}{v} + \frac{3d}{v_1 + 2v_2}}$

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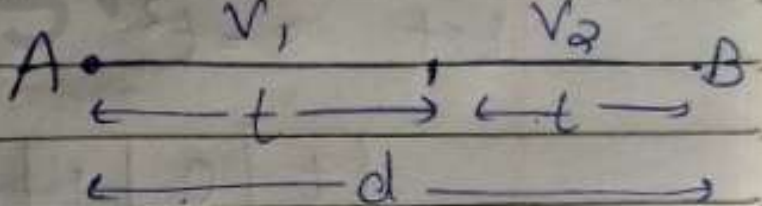
$$= \frac{2}{\frac{1}{v} + \frac{3}{v_1 + 2v_2}}$$

$$= \frac{2}{\frac{v_1 + 2v_2 + 3v}{v(v_1 + 2v_2)}}$$

$$\text{Av. speed} = \frac{2v(v_1 + 2v_2)}{v_1 + 2v_2 + 3v}$$

Ques 6- A man covers a distance with speed v_1 and v_2 for 2 equal time period find av. speed?

Ans 6-



$$d = v_1 t + v_2 t$$

$$2t = \frac{d}{v_1 + v_2}$$

$$\text{Av. speed} = \frac{\text{total distance}}{\text{total time}}$$

$$= \frac{d}{\frac{2d}{v_1 + v_2}}$$

$$\text{Av. speed} = \frac{v_1 + v_2}{2}$$

$$\text{Av. Speed} = \frac{v_1 + v_2}{2}$$

"No religion has mandated killing others as a requirement for its sustenance or promotion." –Dr.A.P.J.Abdul Kalam

Page No.:

Date __/__/____

Instantaneous Velocity and Instantaneous speed.

⇒ Instantaneous velocity

↓

(time should be very small)

$\Delta t \to 0$

↓ dt

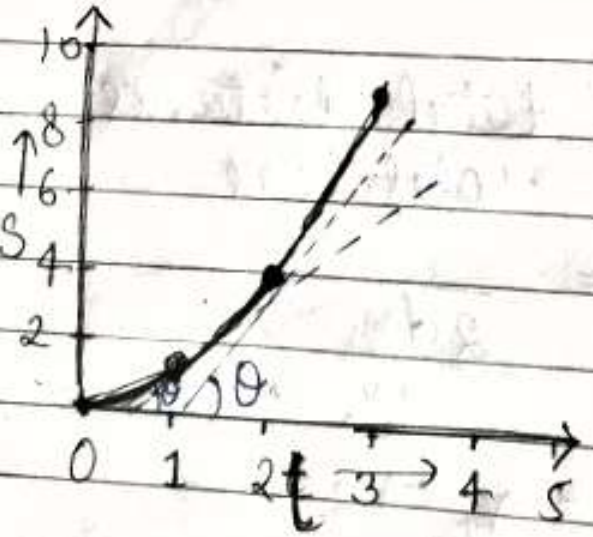
$$v_{inst} = \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

$\frac{ds}{dt}$ ⇒ slope of s v/s t graph (Displacement - time graph)

Let $s = t^2$.

t	0	1	2	3
s	0	1	4	9

Graph



Here, speed is not constant

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Date / /

Page No.

$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{\Delta s}{\Delta t} = \text{velocity}$$

slope of displacement time graph gives velocity

slope is $\tan\theta$

Ques:- Find velocity at $t = 1$ sec and $t = 2$ sec. $t = 3$ sec.

$$s = t^2$$

Ans:- we know that $v = \frac{ds}{dt}$

$$v_{ins} = 2t$$

$$v_{ins} = 2t$$

at $t = 3$ sec

$$v_{inst} = 2 \times 3$$

$$v_{ins} = 6 \text{ m/sec}$$

at $t = 2$ sec

$$v_{inst} = 2 \times 2$$

$$v_{inst} = 4 \text{ m/sec}$$

at $t = 1$ sec

$$v_{inst} = 2 \times 1$$

$$v_{inst} = 2 \text{ m/sec}$$

"...for its sustenance or promotion." -Dr.A.P.J.Abdul Kalam

$V_{average} = \frac{S_2 - S_1}{t_2 - t_1}$

Date __/__/____ Page No.

⇒ What is the average velocity between 1 and 3 sec.

Average velocity = $\frac{\text{Total displacement}}{\text{total time}}$

$= \frac{S_3 - S_1}{2}$

$= \frac{9 - 1}{2}$

Aver. velocity = 4 m/sec.

⇒ At what instant average velocity and instantaneous velocity is same.

At $t = 2$, average velocity and instantaneous velocity is same.

Ques- Given, $S = t^3 - 6t^2$. Find velocity at $t = 1$ sec, $t = 3$ sec, $t = 2$ sec.

Ans- $V_{inst} = \frac{ds}{dt} = 3t^2 - 12t$

At $t = 1$ sec

$V_{inst} = 3 - 12 \Rightarrow -9$ m/sec.

At $t = 3$ sec.

$V_{inst} = 27 - 36 \Rightarrow -9$ m/sec.

At $t = 2$ sec.

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Date / / Page No.

$V_{inst} = 12 - 24 = -12$ m/sec

Ques:- Find the time when the body is at rest

$$S = t^3 - 6t^2$$

$$V_{inst} = \frac{ds}{dt} = 3t^2 - 12t$$

Body is at rest ($v = 0$)

$$3t^2 - 12t = 0$$
$$3t(t - 4) = 0$$
$$3t = 0, \quad t = 4$$
$$t = 0, \quad t = 4$$

Ques:- Given, $s = 3t^2 + 4t + 9$

Find velocity at $t = 2$ sec and average velocity between 1 to 4 sec

Ans:- $V_{inst} = \frac{ds}{dt} = 6t + 4$

at $t = 2$ sec, $12 + 4 = 16$ m/sec.

$$V_{average} = \frac{S_4 - S_1}{t_4 - t_1}$$

$S_4 = 48 + 16 + 9$

$S_1 = 3 + 4 + 9$

"No religion has mandated killing others as a requirement for its sustenance or promotion." –Dr.A.P.J.Abdul Kalam

$v_{average} = \frac{73-16}{3} = \frac{57}{3}$ m/sec.

Acceleration (Rate of change of velocity)

$$\vec{a} = \frac{\vec{v_2} - \vec{v_1}}{\Delta t} = \frac{\Delta \vec{v}}{\Delta t} \quad (m/sec^2)$$

Let $a = 2 m/sec^2$.

v	0	2	4	6
t	0	1	2	3

Average acceleration

$$a_{average} = \frac{\vec{v_2} - \vec{v_1}}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$

Instantaneous acceleration

$$a_{inst} = \frac{dv}{dt}.$$

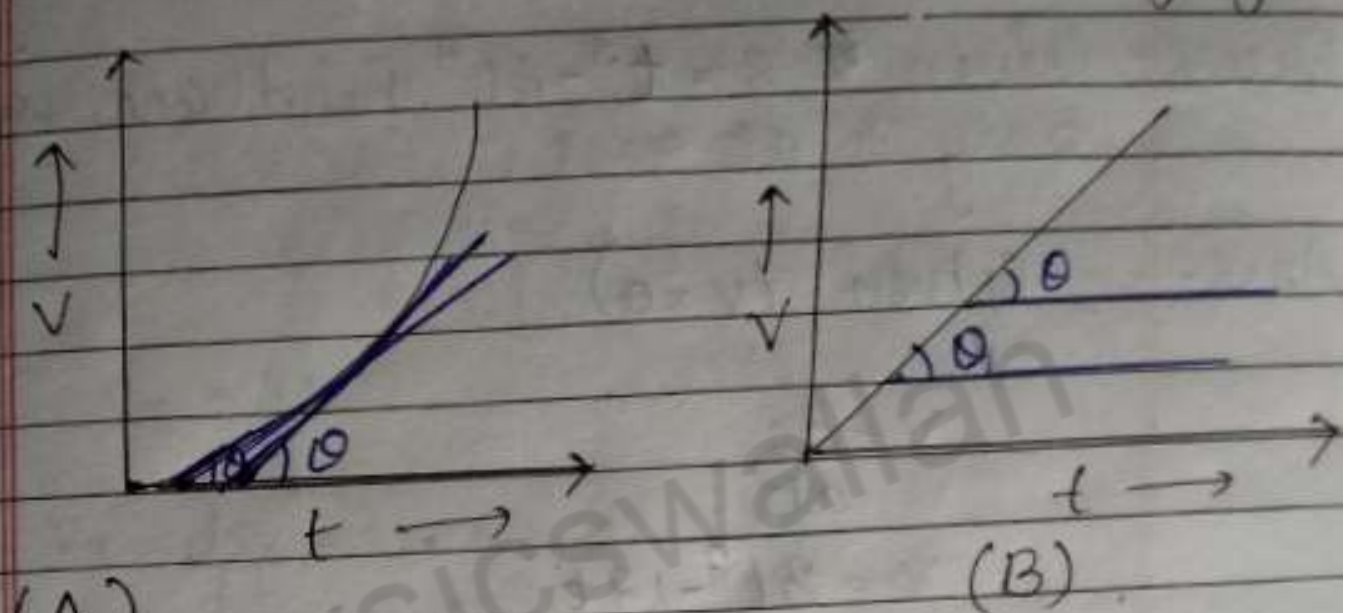
Δt is very small.

v-t graph gives acceleration

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Date / / Page No.

Ques- Consider 2 graph (velocity V/s time) A & B, In which graph the acceleration is constant and in which one acceleration is changing



Ans- In graph A slope is increasing hence the accleration is changing, whereas in graph B, the slope is constant, hence accleration is constant

Ques- Given $v = 2t^2 - 4t$, find acc at t=3sec, and average acc. from 0 to 3 sec.

Ans- $a = \frac{dv}{dt} = 4t - 4$

at $t = 3$ sec

$a = 12 - 4 = 8 m/sec^2$

Page No.

Date ___/___/___

Average acc. from 0 to 3 sec.

$$a = \frac{v_3 - v_0}{3} = \frac{6}{3} = 2 m/sec^2.$$

Ques- Given $s = t^3 - 6t^2$. Find acc. when body is at rest.

Ans- Given, $(v = 0)$

$$v = \frac{ds}{dt}$$

$$v = 3t^2 - 12t$$

$$3t^2 - 12t = 0.$$

$$3t(t - 4) = 0$$

$$t = 0, t = 4$$

$$a = \frac{dv}{dt} = 6t - 12$$

at $t = 0$ sec

$$a = -12 m/sec^2$$

at $t = 4$ sec

$$a = 6 \times 4 - 12$$

$$a = 12 m/sec^2$$

Whenever you take a step forward

Ques: Given -

$s = 3t\hat{i} + 4t\hat{j}$

Find, velocity at 1 sec.
speed at 1 sec.
acc. at 2 sec

Ans:- $\hat{i} \Rightarrow$ displacement in x direction.
$\hat{j} \Rightarrow$ displacement in y direction

$s_x = 3t$ $\qquad$ $s_y = 4t$.

$v_x = \frac{ds_x}{dt} = 3$ $\qquad$ $v_y = \frac{ds_y}{dt} = 4$

$v = 3\hat{i} + 4\hat{j}$

(Here, velocity is constant, it doesn't depend upon time)

(ii) $|\vec{v}| = \sqrt{3^2 + 4^2}$

speed = Magnitude of velocity.

$|\vec{v}| = \sqrt{25}$
$|\vec{v}| = 5 \text{ m/sec}$.

(iii) $a_x = \frac{dv}{dt} = 0$, $a_y = \frac{dv}{dt} = 0$.

$a = 0$.

"No religion has mandated killing others as a requirement for its sustenance or promotion." --Dr.A.P.J.Abdul Kalam

Date __/__/____ Page No.

Ques:- Given $s = t^3\hat{i} + (3t-1)\hat{j}$, find $\vec{v}$, speed, $\vec{a}$ at $t = 1$ sec.

Ans ⇒ $s_x = t^3$ $\quad s_y = 3t - 1$

$v_x = \frac{ds}{dt} = 2t$ $\quad v_y = \frac{ds}{dt} = 3$

$\vec{v} = 2t\hat{i} + 3\hat{j}$

at $t = 1$ sec.

$\vec{v} = 2\hat{i} + 3\hat{j}$

(ii) speed $= |v| = \sqrt{(2)^2 + (3)^2}$

$|v| = \sqrt{4+9}$

$|v| = \sqrt{13}$

(iii) $\vec{a}$ at $t = 1$ sec

$a_x = \frac{dv}{dt} = 2$

$a_y = 0$

$\vec{a} = 2\hat{i}$

$(v>0)$ If velocity is +ive → direction is positive.

$(v<0)$ If velocity is -ive → direction is negative

DIRECTION OF MOTION

Direction of motion is explained by velocity, direction of motion cannot be explained by displacement.

Ques- If $s = t^3 - 6t^2 + 9t$

Find velocity at $t = 2$ sec.

Find acc at $t = 3$ sec.

Ans- $v = \frac{ds}{dt} = 3t^2 - 12t + 9$

$v = 12 - 24 + 9$

$v = -3$ m/sec.

$a = \frac{dv}{dt} = 6t - 12$

at $t = 3$ sec

$a = 18 - 12$

$a = 6$ m/sec^2.

"No nation has prospered at the cost of others as a requirement for its sustenance or promotion." —Dr. A.P.J. Abdul Kalam

Date __/__/____ Page No.

Ques- Find the time at which body is at rest.

$s = t^3 - 6t^2 + 9t$

Ans- $v = 0$

$3t^2 - 12t + 9 = 0$

$t^2 - 4t + 3 = 0$

$t(t-3) - 1(t-3) = 0$

$t = 1, t = 3$

Ques- Find one time when body is moving in negative direction.

$s = t^3 - 6t^2 + 9t$

Ans- $v < 0$.

$v = \frac{ds}{dt} = 3t^2 - 12t + 9$

$3t^2 - 12t + 9 < 0$

$t^2 - 4t + 3 < 0$

$(t-1)(t-3) < 0$

(i) Let $t < 1$

$t = 0$

$(0-1)(0-3)$

$(-1)(-3)$

$\Rightarrow +3$ (Assumption is wrong)

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Date / /

Page No.

(ii) Let $1 < t < 3$

$t = 2$

$(2-1)(2-3)$

$(1)(-1)$

$\Rightarrow -1$

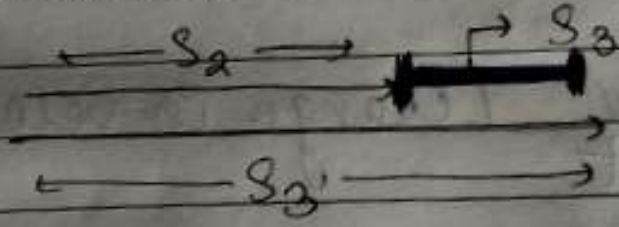
$t = -1$ (Assumption is right)

at $t = -1$, the body is moving in negative direction

At $1 < t < 3$, body is moving in negative direction.

DISPLACEMENT IN n^{th} SECOND

⇒ Displacement in 3rd second



$$S_{3rd} = S_3 - S_2$$

⇒ Displacement in n^{th} second

$$S_{nth} = S_n - S_{n-1}$$

Page No.

Date / /

DERIVATION OF EQUATION OF MOTION UNDER CONSTANT ACCELERATION.

(1) $v = u + at$

(2) $s = ut + \frac{1}{2}at^2$

(3) $v^2 - u^2 + 2as$

where, v = final velocity.
u - initial velocity
a = acceleration
t = time period of acceleration
s = displacement

DERIVATION.

(1) $v = u + at$

$a = \frac{\Delta v}{\Delta t}$ (change in velocity

$a = \frac{dv}{dt}$

$dv = a\,dt$

$\int_u^v dv = \int_0^t a\,dt$

Date / / Page No.

$$[v]_u^v = a[t]_0^t$$

$$[v-u] = a[t-0]$$

$$v-u = at$$

$$\boxed{v = u+at}$$

(2) $s = ut + \frac{1}{2}at^2$.

$$v = \frac{ds}{dt}$$

$$ds = v\,dt$$

$$\int ds = \int v\,dt = \int (u+at)\,dt$$

$$\int_0^s ds = \int_0^t (u+at)\,dt$$

$$[s]_0^s = u\int_0^t dt + a\int_0^t t\,dt$$

$$[s]_0^s = u[t]_0^t + a\frac{t^2}{2}$$

$$\boxed{s = ut + \frac{1}{2}at^2}$$

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"No religion has mandated killing others as a requirement for its sustenance or promotion." –Dr.A.P.J.Abdul Kalam

Date __/__/____ Page No..

(3) $v^2 - u^2 = 2as$

$$a = \frac{dv}{dt}$$

$$a = \frac{dv}{dt} \times \frac{ds}{ds}$$

$$a = \frac{dv}{ds} \times \frac{ds}{dt}$$

$$a = \left(\frac{dv}{ds}\right) v$$

$$a = v \frac{dv}{ds}$$

$$a\,ds = v\,dv$$

$$\int_0^s a\,ds = \int_u^v v\,dv$$

$$a[s]_0^s = \left[\frac{v^2}{2}\right]_u^v$$

$$as = \frac{v^2 - u^2}{2}$$

$$\boxed{v^2 - u^2 = 2as}$$

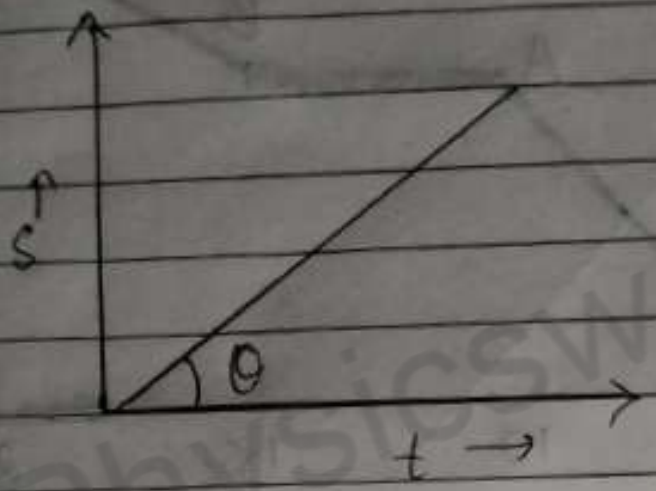
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Date / / Page No.

GRAPH

⇒ Displacement - time Graph, Velocity - time graph, Acceleration - time graph

⇒ Displacement - time Graph



The angle made by graph with positive x-axis, denotes slope.

$$\theta \rightarrow \text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{s_2 - s_1}{t_2 - t_1}$$

$$= \frac{\Delta s}{\Delta t} \rightarrow \text{velocity}$$

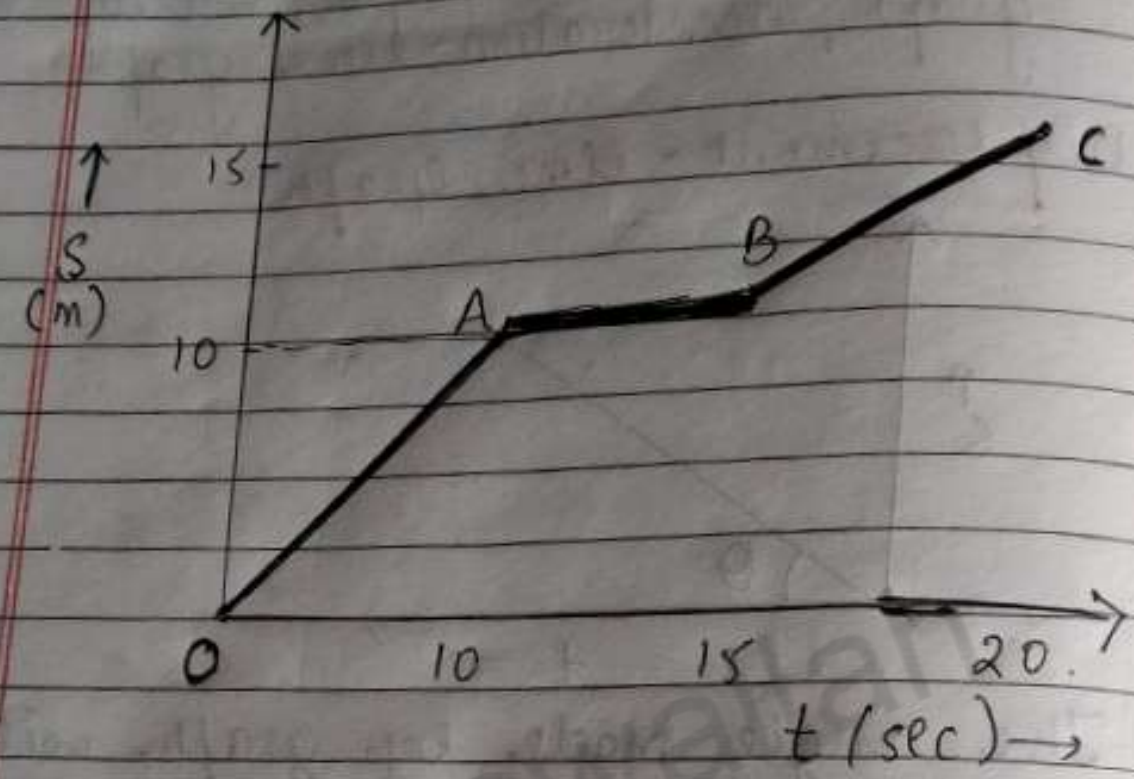
θ increases, velocity increases

slope = $\tan\theta$ = velocity

Page No.

Date / /

Ques- Find velocity in region OA, AB, BC?



Ans- ⇒ Velocity in OA Region

$$V \rightarrow \text{slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$V_{OA} = \frac{10 - 0}{10 - 0}$$

$$V_{OA} = \frac{10}{10} = 1 \text{ m/sec}$$

⇒ Velocity in AB Region,

Slope is zero, so $V_{AB} = 0$ m/sec

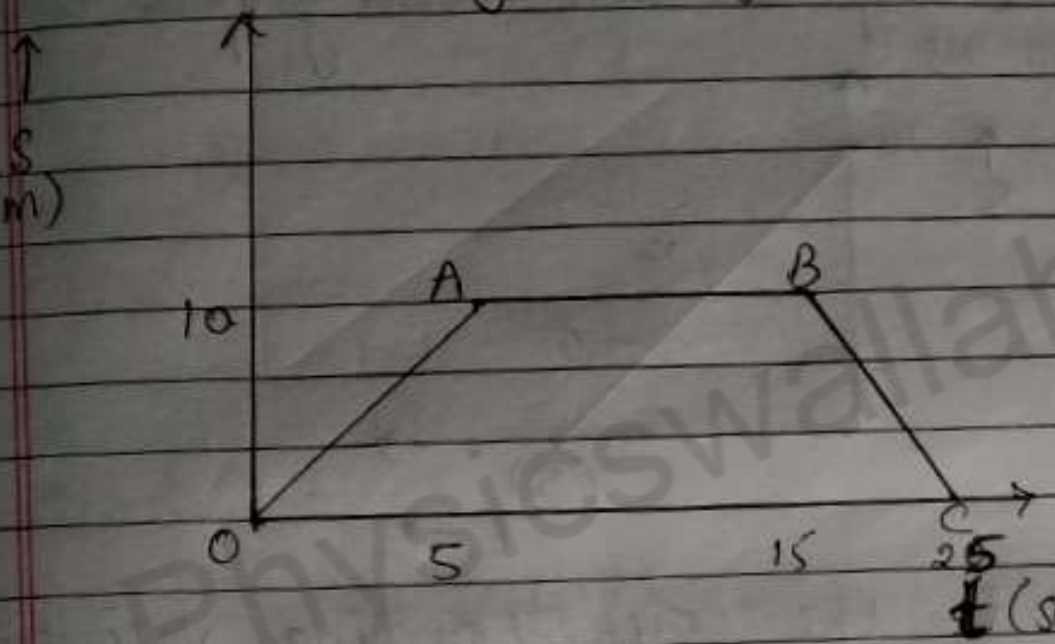
"Whenever you take a step

Date ___/___/___ Page No.: ___

⇒ Velocity in BC region

$$V_{BC} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{15-10}{20-15} = 1 \text{ m/sec}$$

Ques- Find velocity in region OA, AB, BC ?



Ans- Velocity in OA region

$$V_{OA} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{10-0}{5-0} = \frac{10}{5} = 2 \text{ m/sec}$$

Velocity in AB region

$$V_{AB} = 0 \text{ m/sec}$$

Velocity in BC region

$$V_{BC} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0-10}{25-15} = \frac{-10}{10} = -1 \text{ m/sec}$$

"No religion has mandated knowledge as a requirement for its sustenance or promotion." - Dr. A.P.J. Abdul Kalam

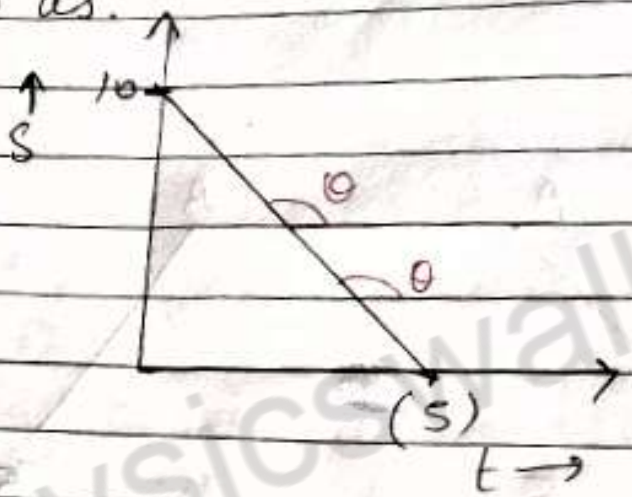
Page No..

Date __/__/____

* If $\theta < 90^\circ$ (with positive x-axis) slope is positive

* If $\theta > 90^\circ$ (with positive x-axis) slope is negative

Ques- The Displacement-time graph is given as.



Find the corresponding velocity-time graph.

Ans- slope is negative.

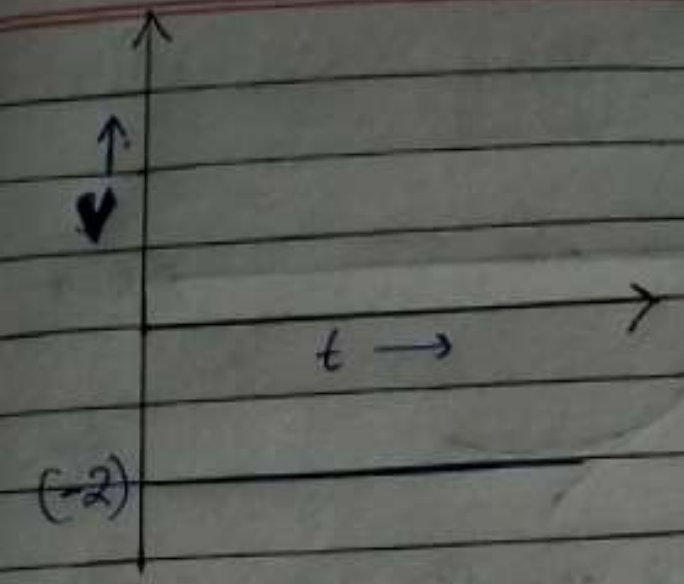
$\theta \rightarrow$ constant.

$v \rightarrow$ constant.

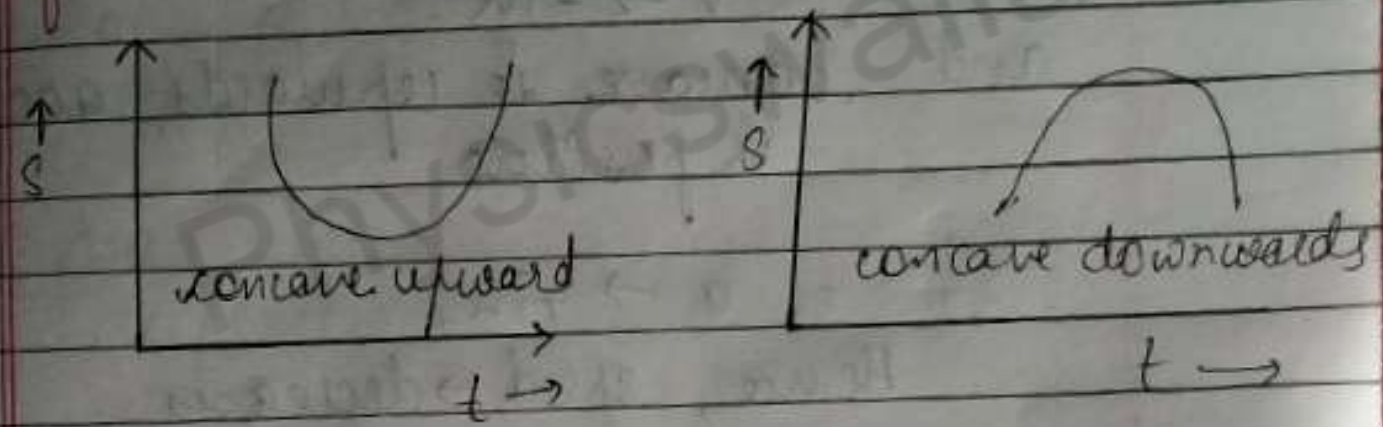
$$v = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 10}{5 - 0} = \frac{-10}{5}$$

$$v = -2 \text{ m/sec}$$

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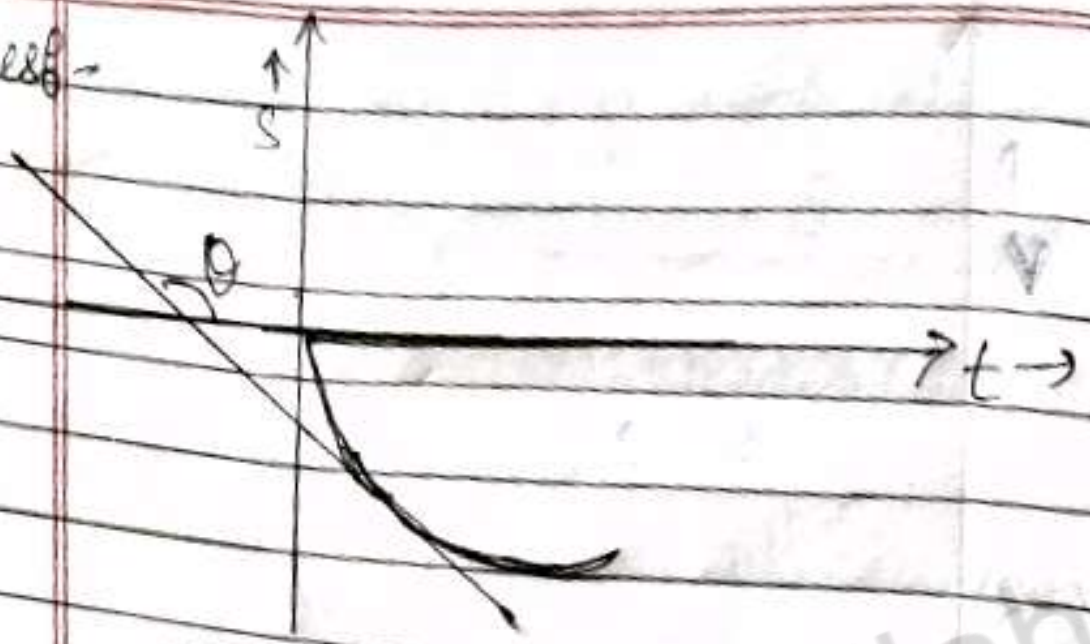
* If concave upward, acceleration positive
* If concave downwards, acceleration negative



* If v and a has same sign ~~velocity~~ speed increases

* If v and a has opposite sign, ~~velocity~~ speed decreases

Date __/__/____ Page No..

Ques- 

Find v, a, speed?

Ans- Here θ is greater than 90°, which means

$v \rightarrow -ive$.

and concave is upwards, acc. is positive

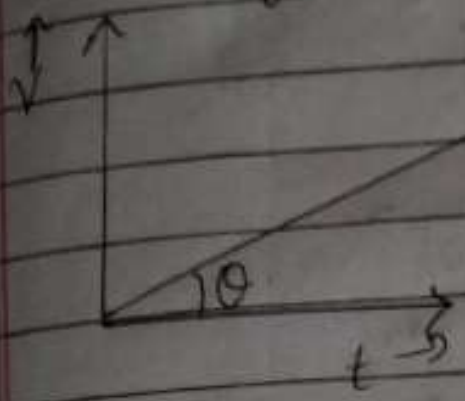
$a \rightarrow +ive$.

Hence, speed → decreases

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Date / / Page No.

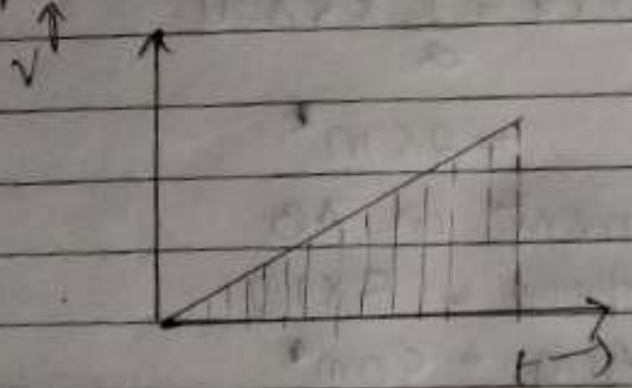
⇒ Velocity - time graph :



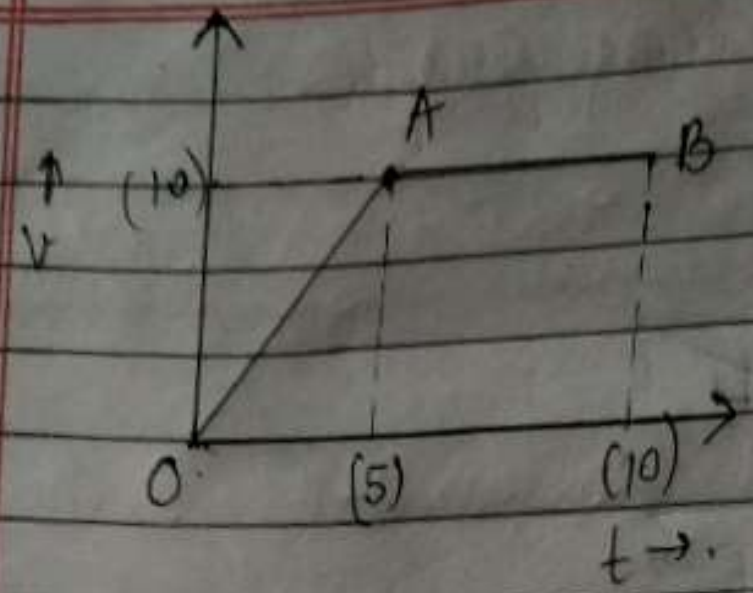
$\theta \rightarrow \text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t} = a$

slope of v-t graph represent acceleration

* Area under v-t curve represent displacement. (area with time axis)



Ques- Given v-t graph, find acc and displacement of the graph.



Ans6- Acceleration in OA region

$$\text{slope} = \frac{10-0}{5-0}$$

$$\text{slope} = 2 \text{ m/sec}^2$$

Acceleration in AB region is zero.

⇒ Displacement in OA

$$\text{Area} = \frac{1}{2} \times 5 \times 10.$$

$$= 25 \text{ m}$$

⇒ Displacement in AB

$$\text{Area} = 5 \times 10$$

$$\text{Area} = 50 \text{ m}$$

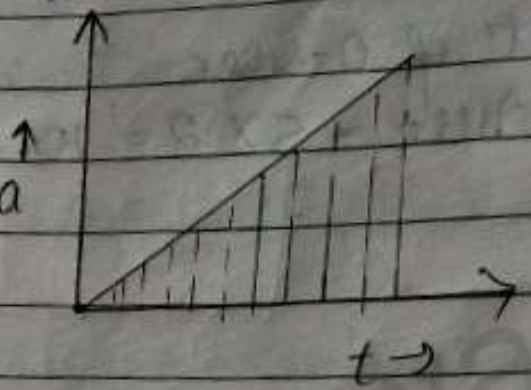
Page No.

Date / /

Acc - time Graph.

Slope $= \frac{\Delta a}{\Delta t} =$ Jerk, m/sec^3

Area of a-t curve = change in velocity



MOTION UNDER GRAVITY

⇒ Motion under gravity is example of uniform acceleration

⇒ $a = g \leftarrow 9.8$ or 10

For uniform acceleration we know that

⇒ $s = ut + \frac{1}{2}at^2$

⇒ $v^2 - u^2 = 2as$

⇒ $v = u + at$

Date / /

Page No.

CONVENTION

When the object is moving upward
⇒ the velocity is positive.
⇒ $a = -g$
⇒ s = +ive

When the object is moving downwards
⇒ the velocity is taken as negative.
⇒ $a = -g$
⇒ s = -ive

Ques: A body is thrown vertically upward with speed 'u'. Find
(i) time taken by it to reach Max. ht.
(ii) Max. height.
(iii) time taken to reach initial position

Max. height • ($v = 0$)

$a = -g$ ↑ $u = +u$

(i) $v = u + at$
$0 = u + (-gt)$
$u = gt$
$t = \frac{u}{g}$

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Page No.

Date / /

(ii) Let the max. height be 'h'

$s = +h$

$v^2 = u^2 + 2as$

$0 = u^2 + 2(-g)(h)$

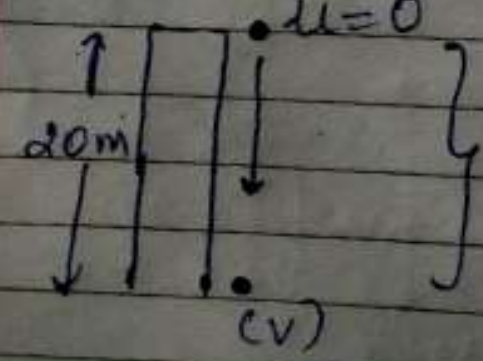
$u^2 = 2gh$

$$\boxed{h = \frac{u^2}{2g}}$$

(iii) Total time $= 2t$

Total time $= 2\left(\frac{u}{g}\right)$

Ques- A stone is dropped from a tower of height 20m. Find
(i) time taken to reach ground.
(ii) speed just before hitting ground.



$a = -g$

$u = 0$
$a = -g$
$s = -20$
$t = ?$

Date / /

Page No.

$$s = ut + \frac{1}{2}at^2.$$

$$-20 = 0 \times t + \frac{1}{2}(-10) \times t^2.$$

$$-20 = -5t^2.$$

$$4 = t^2$$

$$t = 2 \text{ sec.}$$

(ii) $v = u + at$

$v = 0 - gt$

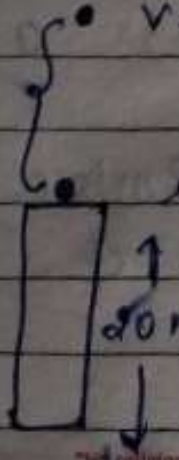
$v = -10 \times 2$

$v = -20 \text{ m/sec}$

Ques- A ball is projected vertically upwards from a tower of height 20m at a speed of 5m/sec. Find.

(i) Max. height reached by ball (from ground)
(ii) time when it passes from its mean position

Ans:-



$u = +5 \text{ m/sec}$

$a = -g$

$v = 0$

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Page No.

Date ___/___/___

$v^2 = u^2 + 2as$

$0 = (5)^2 + 2(-10)s$

$20s = 25$

$s = \frac{25}{20} = 1.25\ m$

(Vapis usi jagah)

(ii)



$v = u + at$

$0 = 5 - 10 \times t$

$t = 0.5\ sec$

time when it passes from its mean position $= 2t$

$= 1\ sec$

Ques:- A ball is thrown upwards at a speed of 28 m/sec. Find the speed of ball 1 sec before reaching maximum height.

Ans:-



At maximum height $v = 0$.

$v = u + at$

"Whenever you take a step forward

Date / / Page No.

$v = u + at$

$0 = 28 + (-10)t$

$t = 2.8 \text{ sec}$

At 2.8 sec the ball will reach its max height.

Before 1 sec i.e 1.8 sec

$u = 28 \text{ m/sec}$

$a = -10$

$t = 1.8$

$v = u + at$

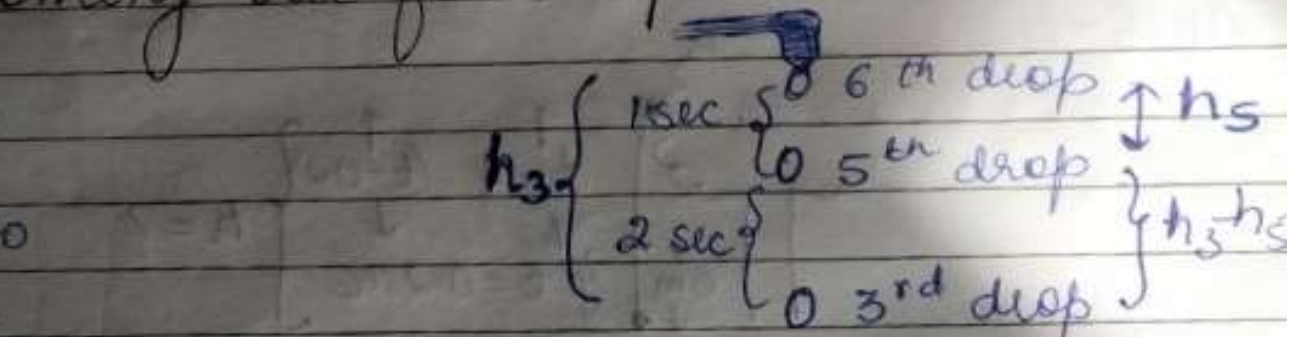
$v = 28 - 10 \times (1.8)$

$28 - 18$

$v = 28 - 18$

$v = 10 \text{ m/sec}$

Ques6- A tap drops water drops at one second intervals. Find the distance between 3rd and 5th drop when 6th drop is just coming out from tap.



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Date __/__/____ Page No..

$$S = ut + \frac{1}{2}at^2$$

$$= 0 + \frac{1}{2} \times -10 \times 9$$

$$S = -45\,m$$

5th drop

$u = 0$

$a = -10$

$t = 1\,sec$

$$S = ut + \frac{1}{2}at^2$$

$$= 0 + \frac{1}{2} \times (-10) \times 1$$

$$S = -5\,m$$

Distance between them is 40 m.

Date __/__/____

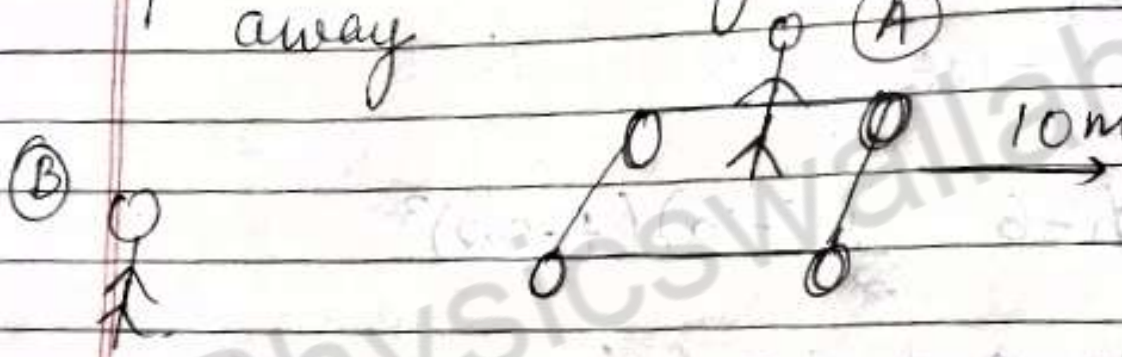
RELATIVE VELOCITY / ACCELERATION / DISPLACEMENT

⇒ These all are described by Frame of reference.

⇓

Position of observer.

Consider a person A in trolley moving with the speed of 10 m/sec. Another person B watching A at some distance away.



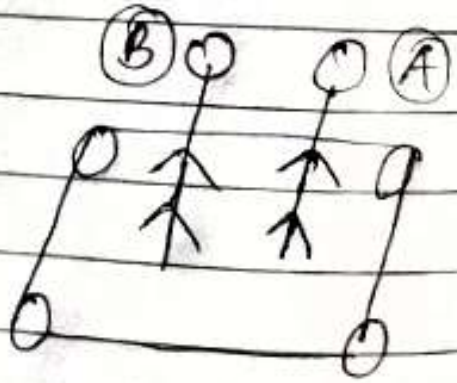
$$V_{AB} = \vec{V}_A - \vec{V}_B$$

$$V_{AB} = 10 - 0$$

$$V_{AB} = 10 \text{ m/sec}$$

(Velocity of A w.r.t B)

Now, when B joins A



Velocity of A w.r.t B

$$V_{AB} = \vec{V}_A - \vec{V}_B$$

$$V_{AB} = 10 - 10$$

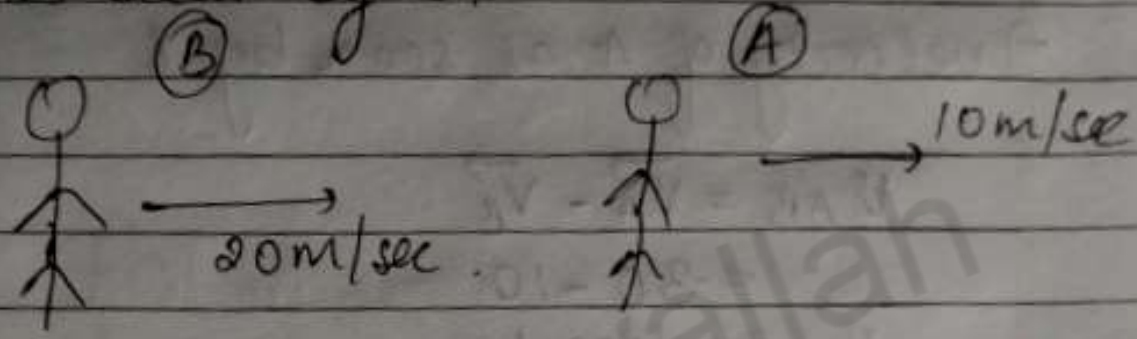
$$V_{AB} = 0 \text{ M/sec}$$

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Date / / Page No.:

* Time doesnot depends upon Frame of Reference.

Quesb- consider 2 people A & B, A is travelling with the speed of 10m/sec, and B is travelling at 2m/sec when B observes A, what is the speed of A as seen by B.



Ansb- $V_{AB} = \vec{V}_{AG} - \vec{V}_{BG}$

$V_{AB} = 10 - 20$

$V_{AB} = -10 m/sec.$

⇒ speed of B as seen by A

$V_{BA} = \vec{V}_{BG} - \vec{V}_{AG}$

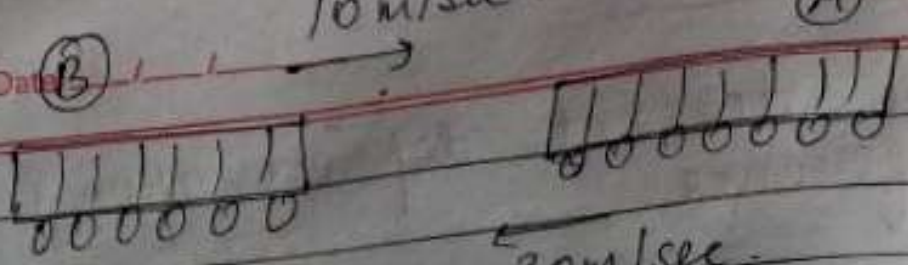
$V_{BA} = 20 - 10$

$V_{BA} = 10 m/sec.$

Quesb- consider 2 travelling travelling off to each other on same road A by 30m/sec, B by 10m/sec, what is the velocity of B as seen by A.

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Page No.
Date / /



Ans8- $v_{BA} = \vec{v_B} - \vec{v_A}$

$= 10 - (-30)$

$v_{BA} = 40 \text{ m/sec}$

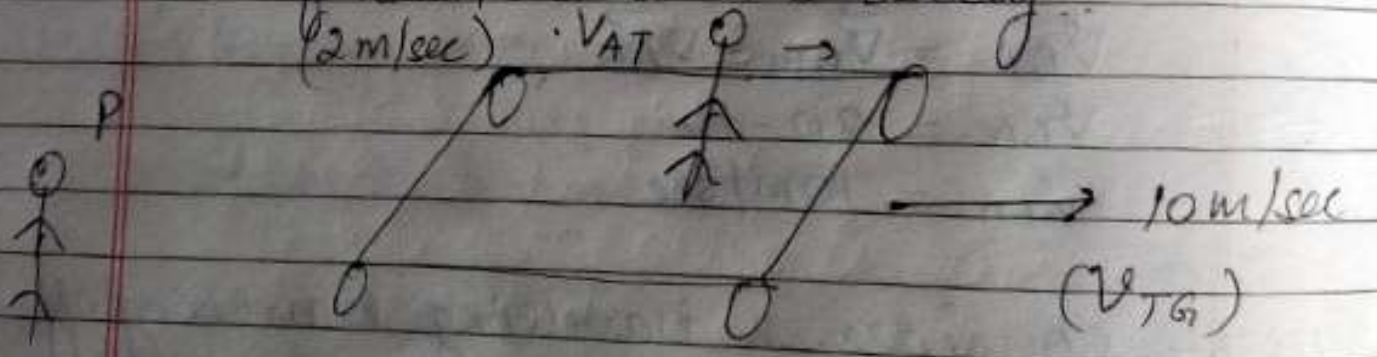
⇒ velocity of A as seen by B

$v_{AB} = \vec{v_A} - \vec{v_B}$

$= -30 - 10$

$v_{AB} = -40 \text{ m/sec}$

Ques8- what is the velocity of A w.r.t P. as, the ~~speed~~ velocity of trolley w.r.t ground is 10m/sec, and a object A is moving at the speed of 2 m/sec w.r.t trolley.



Ans8- $v_{AP} = \vec{v_{AG}} - \vec{v_{PG}}$

$\vec{v_{AG}} = ?$

"Whenever you take a step forward, you are bound to disturb something." —Indira Gandhi

$\vec{V}_{AT} = \vec{V}_{AG} - \vec{V}_{TG}$

$2 = \vec{V}_{AG} - 10$

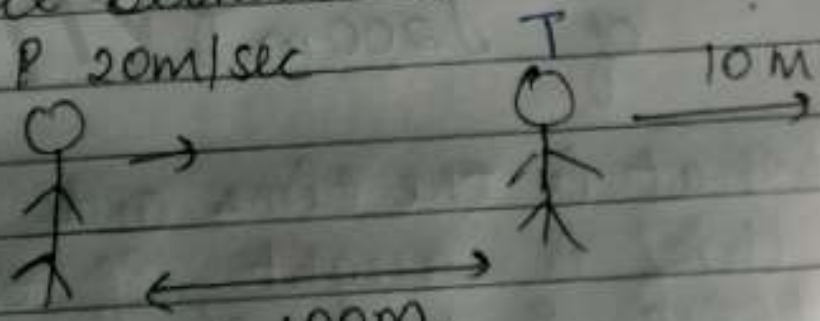
$\vec{V}_{AG} = 12$ m/sec.

$\vec{V}_{AP} = 12 - 0$.

$\vec{V}_{AP} = 12$ m/sec.

POLICE-THIEF PROBLEM.

Ques- Find the time at which police caught the thief. If speed of Thief is 10m/sec and police is 20m/sec and the distance between them is 100m.



Ans- Suppose T is at rest

$V_{PC} = V_{PG} - V_{TG}$

$= 20 - 10$

$V_{PC} = 10$ m/sec.

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Page No.:

Date __/__/____

S_{PT} (Relative distance between Police and Thief)

$S_{PT} = 100$.

⇒ Relative distance does not depend upon frame of reference

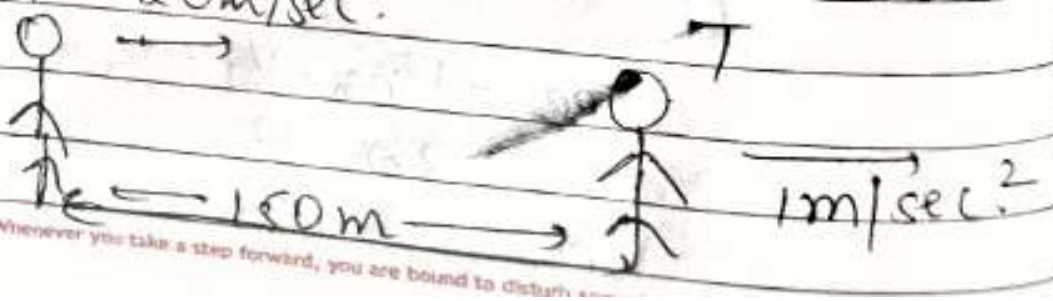
$$t = \frac{S}{v} = \frac{100}{10} = 10 \text{ sec}$$

Discovered by thief in 10 sec.

$$S = v \times t$$
$$S = 10 \times 10$$
$$S = 100 \text{ m},$$

Police caught thief at a distance of 200 m.

Ques- What is the time and position when thief is caught, If the acc of thief is $1 m/sec^2$, and the speed of police is 20 m/sec, and distance between them is 150 m



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"Whenever you take a step forward, you are bound to distan

Date / / Page No.

Ans 6- $v_{pc} = v_p - v_c$

$= 20 - 0$

$v_{pc} = 0$

$a_{pc} = a_p - a_c$

$= 0 - 1$

$= -1 \text{ m/sec}^2$.

$s_{pc} = 150\text{m}$

$s = ut + \frac{1}{2}at^2$

$150 = 20t + \frac{1}{2}(-1)t^2$.

$t^2 - 40t + 300 = 0$.

$t^2 - 30t - 10t + 300 = 0$.

$(t-30)(t-10) = 0$

$t = 30\text{sec}, t = 10\text{sec}$

⇒ After 10 sec. from (Police)

$s = 20 \times 10 = 200\text{m}$.

from

⇒ after Thief

$s = ut + \frac{1}{2}at^2$

$s = 0 \times t + \frac{1}{2} \times (1) \times 100$.

$s = 50\text{m}$.

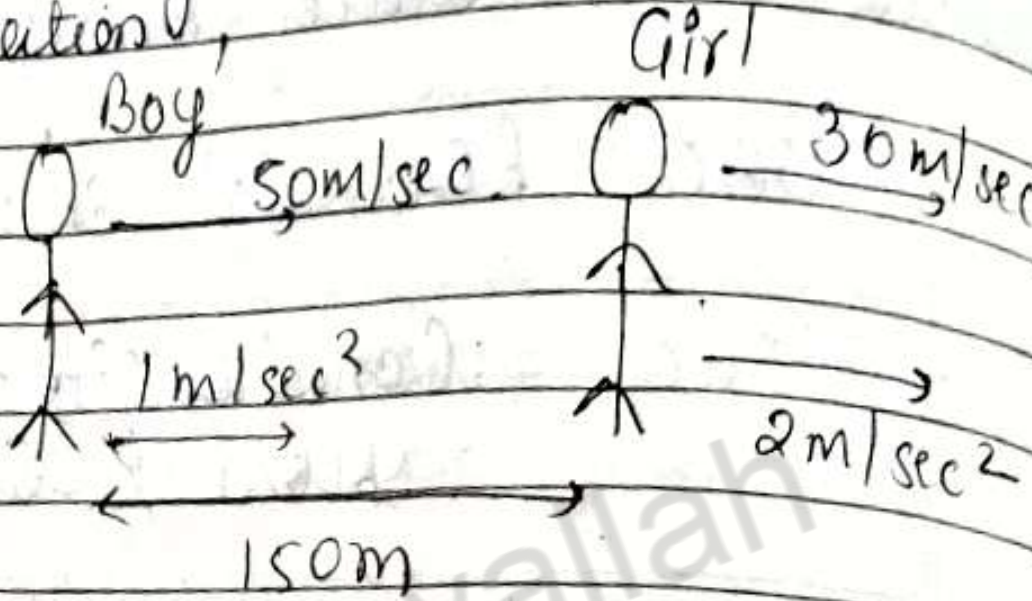
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Page No.:

Date __/__/____

Ques- Find the time when boy catches the girl, or they are at their closest separation,



$$V_{BG'} = V_{BG} - V_{G'G}$$
$$= 50 - 30 = 20 \text{ m/sec}$$

$$a_{BG'} = a_{BG} - a_{G'G}$$
$$= 1 - 2 = -1 \text{ m/sec}^2$$

$$S_{BG'} = 150 \text{ m}$$

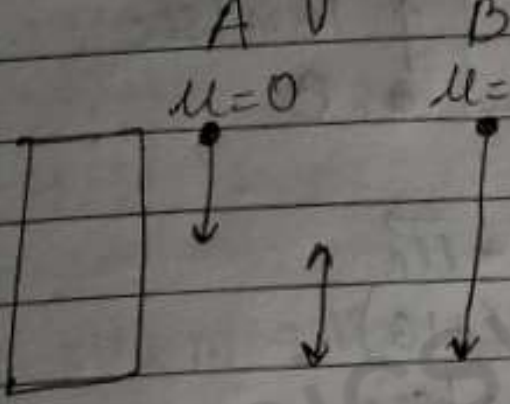
$$S = ut + \frac{1}{2}at^2$$

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Date __/__/____ Page No.: ____

MOTION UNDER GRAVITY

Ques- Find the separation Between A and B after 3 sec, If 2 ball with initial speed 0 & 3m/sec respectively are thrown from building



$V_{AB} = V_A - V_B$

$= 0 - (-3)$

$V_{AB} = +3$

$a_{AB} = a_A - a_B$

$= -g - (-g)$

$a_{AB} = 0$

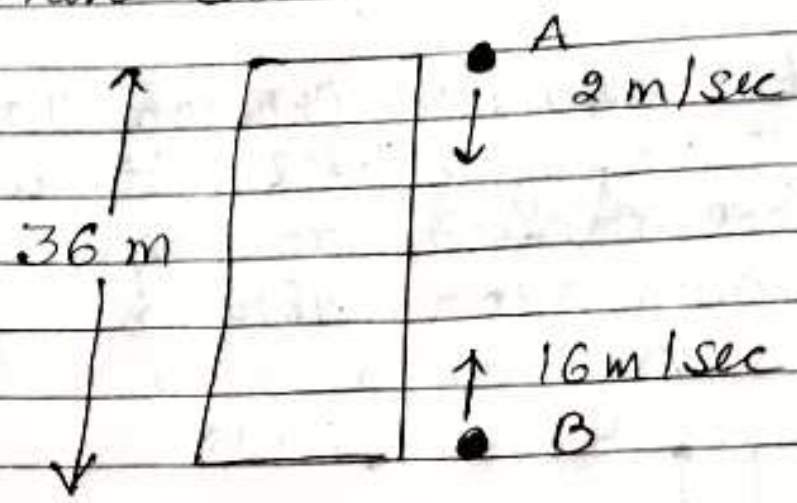
$t = 3$ sec.

$S = vxt$

$\boxed{S = 9m}$ Separation.

Date __/__/____ Page No.

Ques6- When the two ball will meet.



Ans6- $u_{AB} = \vec{u_A} - \vec{u_B}$

$= -2 - (+16)$

$u_{AB} = -18 \text{ m/sec}$

$a_{AB} = -g - (-g)$

$a_{AB} = 0$

$S_{AB} = 36 \text{ m}$

$t = \frac{S}{u} = \frac{36}{18} = 2 \text{ sec}$

They will meet after 2 sec.

⇒ The distance travelled by B is

$$S = ut + \frac{1}{2}at^2$$

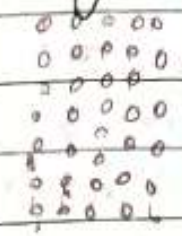
$$S = 16 \times 2 + \frac{1}{2} \times (-10) \times 2^2$$

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RAIN-MAN PROBLEM.

Ques- It's raining at the speed of 40 km/hr. What is the velocity of rain as seen by man?

Ans-



$$v_{RM} = v_{RG} - v_{MG}$$

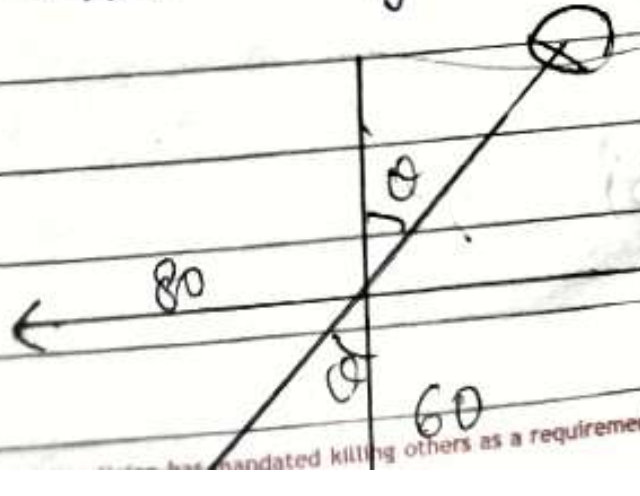
$$v_{RM} = -40\hat{j} - 0$$

$$v_{RM} = -40\hat{j}.$$

Ques- It is raining at the rate of 60 km/hr. and a man is moving in trolley at the rate of 80 km/hr. at what angle, the man will open its umbrella.

$$v_{RM} = v_{RG} - v_{MG}$$

$$v_{RM} = -60\hat{j} - 80\hat{i}$$



$$\tan\theta = \frac{80}{60} = \frac{4}{3}$$

$$\theta = 53°.$$

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mandated killing others as a requirement for its sustenance or promotion." —Dr. A. P. J. Abdul Kalam

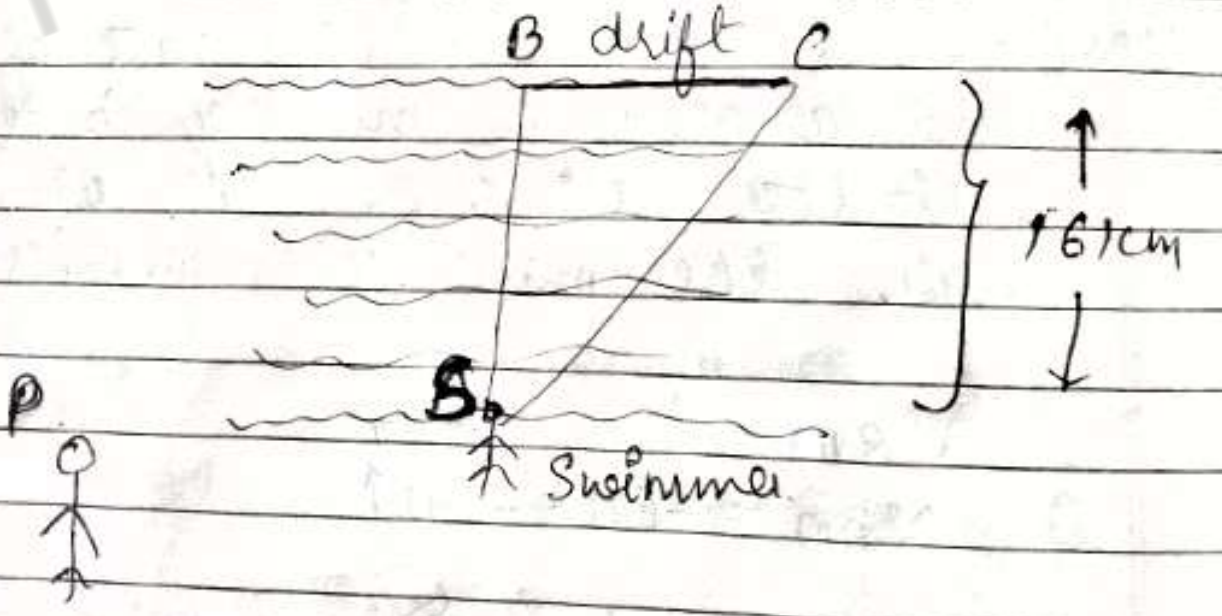
Date __/__/____ Page No.__

RIVER SWIMMER / RIVER BOAT / AEROPLANE WIND

Ques- The river is flowing at the rate of 3 km/hr, a man tries to flow perpendicular to the flow of river at the velocity of 4 km/hr. According to the river V_{SR} (velocity of swimmer acc to river) is in $\hat{i}$ direction. The person outside the river (P) saw that the river drift's the swimmer. The width of swimmer is given as 16 km. Find

(1) time taken by swimmer to cross river.

(2) At what point on the opposite of bank did the swimmer arrived.



Ans- $V_{SG} = 4\hat{j} + 3\hat{i}$

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Date __/__/____ Page No.. ____

$V_{SG} = V_{SR} + V_{RG}$

(The swimmer got 2 velocities)

Vertical motion

$S = 16$ km

$V = 4$ km/hr

$t = \frac{16}{4} \Rightarrow 4$ hr

Drift is in x-direction.

Horizontal

$d = s \times t$

$d = 3$ km/hr $\times 4$ hr

$d = 12$ km (drift).

Actually distance he swims $= \sqrt{(16)^2 + (12)^2}$

$= 20$ km.

Vertical motion will be covered by vertical speed, horizontal motion will be covered by horizontal speed.

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Page No.

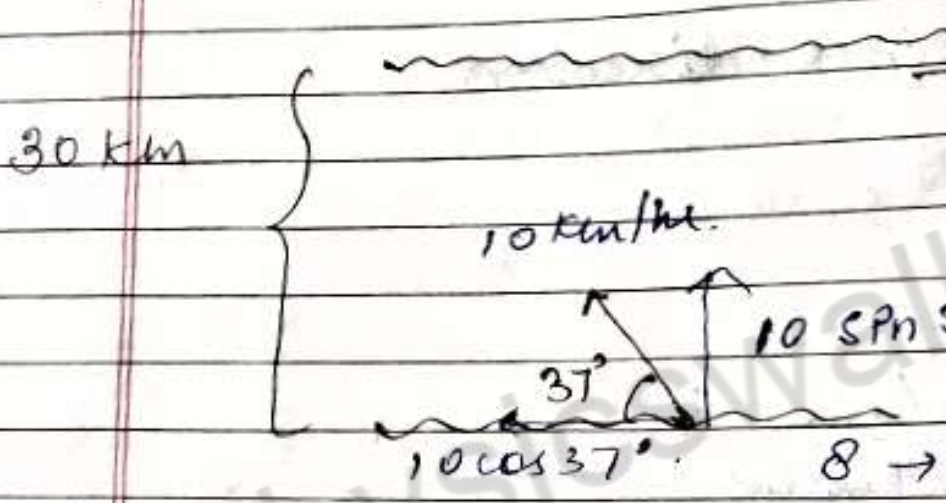
Date __/__/____

Ques6- A river flows at an rate of 8 km/hr (V_R), width of river is 30km, A man swims at an angle of 37° at a speed of 10 km/hr. Find

(i) time taken to cross river.

(ii) At what point does he arrives (deft).

(iii) actual distance he swims.



Ans6- $10\cos 37° = 8$

$10\sin 37° = 6$.

velocity of swimmer w.r.t ground = velocity of swimmer w.r.t river + velocity of river w.r.t ground

$$V_{SG} = V_{SR} + V_{RG}$$

Horizontal speed = 0

$V_{SG} = 6\uparrow$

(i) $t = \frac{30}{6} = 5$ hour.

"Whenever you take a step forward, you are bound to disturb something." —Indira Gandhi

Date __/__/__

Page No.: _____

(ii) vertically opposite. (He crosses vertically opposite)

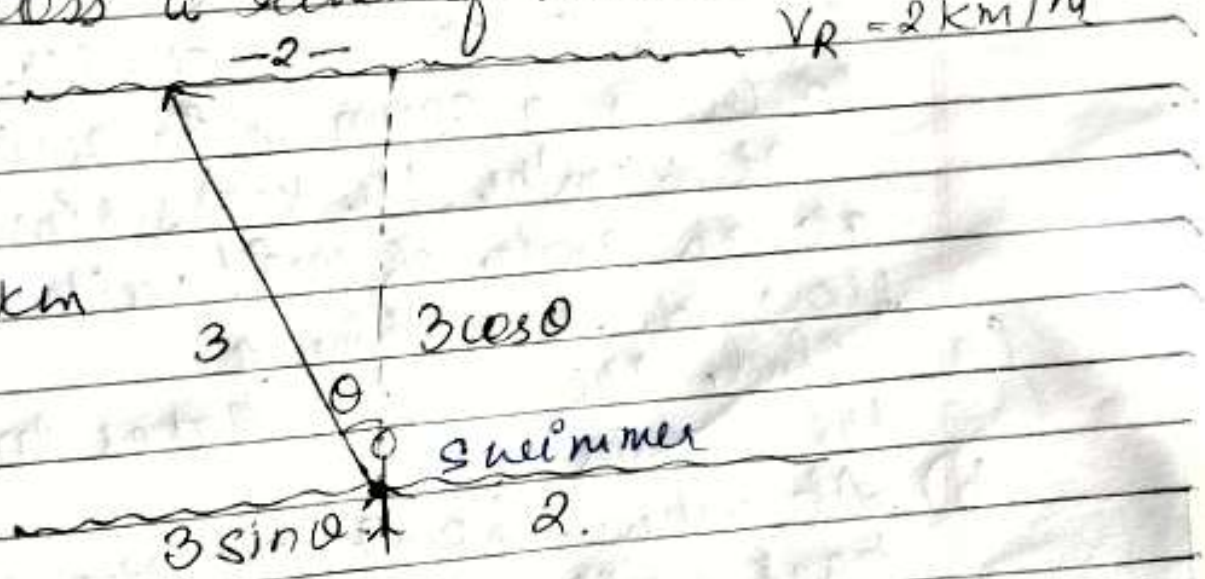
i.e drift = 0

(iii) Actual distance = 30 km.

Ques6- A man can swim in still water at a speed of 3 km/hr. He wants to cross a river that flows at 2 km/hr and reaches the point directly opp. to his starting point.

(i) In which direction should he try to swim.

(ii) How much time he will take to cross a river of width 500m wide.



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"...others as a requirement for its sustenance or promotion." —Dr.A.P.J.Abdul Kalam

Date __/__/ $V_{SG} = V_{SR} + V_{RG}$ Page No.

Ans:- To reach the point directly opp to his starting point, the man should swim with some angle w.r.t its vertical position,

There should be no horizontal velocity so,

$$3\sin\theta = 2$$
$$\sin\theta = 2/3$$

(ii) $t = \frac{0.5}{3\cos\theta}$

$\cos\theta = \sqrt{1-\sin^2\theta}$

$\cos\theta = \frac{\sqrt{5}}{3}$

$t = \frac{1}{2\sqrt{5}}$ hours

Ques6- A man can swim at a speed of 3 km/hr in still water. He wants to cross a 500m wide river flowing at 2 km/hr. He keeps himself always at an angle of 120° with the river flow while swimming.

(a) Find the time he takes to cross the river.

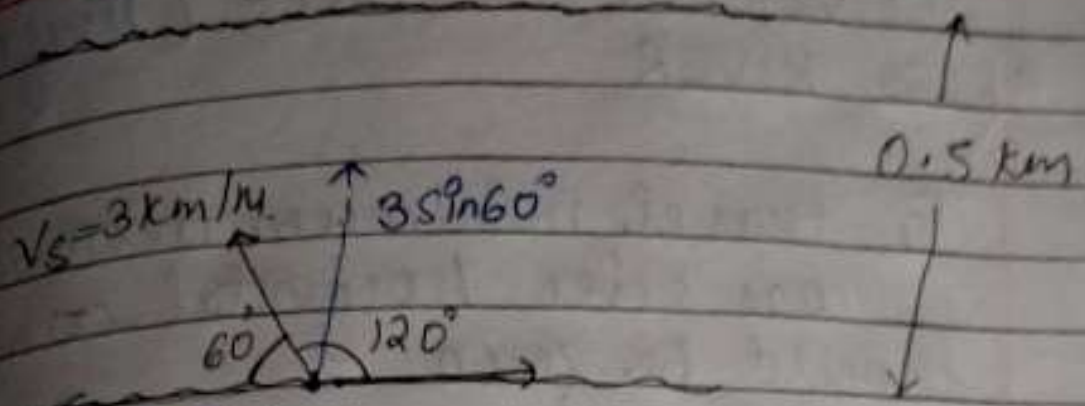
(b) At which point on the opposite bank will he arrive?

Physicswallah

Date / /

Page No.

$V_R = 2 km/hr$



$3 \cos 60° = 1.5$

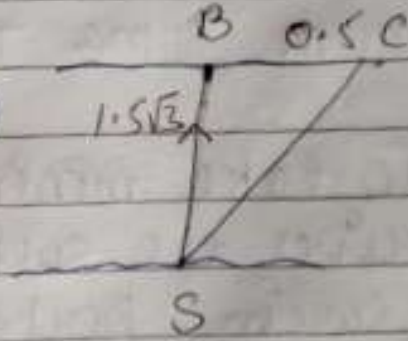
$3 \sin 60° = 1.5\sqrt{3}$

$V_{SG} = V_{SR} + V_{RG}$

$V_{SG} = 0.5\hat{i} + 1.5\sqrt{3}\hat{j}$

$t = \frac{0.5}{1.5\sqrt{3}}$ (Vertical direction)

$t = \frac{1}{3\sqrt{3}}$



(ii) The swimmer will go towards C

$BC = \text{speed} \times t$

$= 0.5 \times \frac{1}{3\sqrt{3}} = \frac{1}{6\sqrt{3}}$ km (Horizontal)

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Page No.:

Date __/__/____

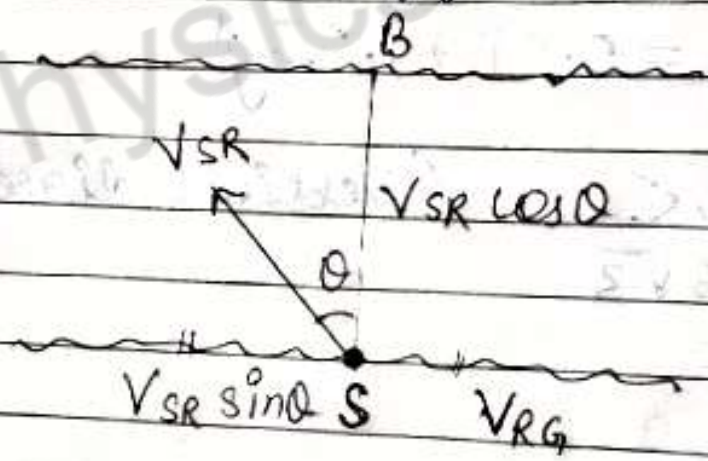
MINIMUM DISTANCE (Drift = 0) TO CROSS RIVER

To travel minimum distance to cross river horizontal component should be zero.

$$V_{SR} \sin\theta = V_{RG}$$

$$\boxed{\sin\theta = \frac{V_{RG}}{V_{SR}}}$$

As shown in figure



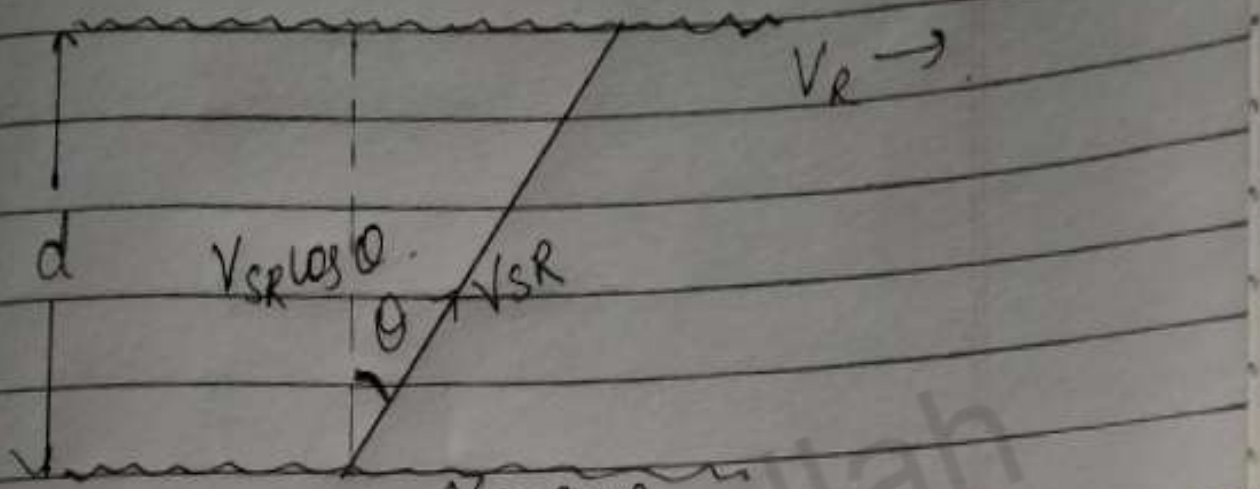
MINIMUM TIME TO CROSS RIVER

To take minimum time to cross river the swimmer should try to swim perpendicular to river flow.

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Date / / Page No.:

If the swimmer swims at some angle θ.



$$t = \frac{d}{V_{SR}\cos\theta} \quad \text{(vertical component)}$$

for $t \rightarrow$ min

$\cos\theta \rightarrow$ max

Max. value of $\cos\theta = 1$

and $\theta = 0^\circ$

so to take minimum time to cross river the swimmer should try to swim $\perp$ to river flow.

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